

CS2601 Linear and Convex Optimization

Homework 9 Solution

Due: 2021.12.2

1. Consider the equality constrained quadratic program

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) = \frac{1}{2}x_1^2 + x_1x_2 + x_2^2 - x_1 - 3x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 = 1 \end{aligned}$$

- (a). Find the optimal solution x_1^*, x_2^* by reduction to an unconstrained problem.
- (b). Find the optimal solution and the corresponding Lagrange multiplier λ^* using the Lagrangian multipliers method.

Solution. (a). Eliminating x_1 using $x_1 = 1 - 2x_2$, we obtain an unconstrained problem

$$g(x_2) = f(1 - 2x_2, x_2) = x_2^2 - 2x_2 - \frac{1}{2}$$

Minimizing g ,

$$x_2^* = 1$$

and

$$x_1^* = 1 - 2x_2 = -1$$

- (b). The Lagrangian is

$$\mathcal{L}(x_1, x_2, \lambda) = \frac{1}{2}x_1^2 + x_1x_2 + x_2^2 - x_1 - 3x_2 + \lambda(x_1 + 2x_2 - 1)$$

The Lagrange condition is

$$\partial_{x_1}\mathcal{L} = x_1 + x_2 - 1 + \lambda = 0$$

$$\partial_{x_2}\mathcal{L} = x_1 + 2x_2 - 3 + 2\lambda = 0$$

$$\partial_{\lambda}\mathcal{L} = x_1 + 2x_2 - 1 = 0$$

Solving the linear system yields

$$x_1^* = -1, \quad x_2^* = 1, \quad \lambda^* = 1$$

The problem is convex, since

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \succ \mathbf{O}$$

So what we find is the optimal solution.

□

2. Consider the equality constrained quadratic program

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{g}^T \mathbf{x} + c \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \end{aligned}$$

where $\mathbf{Q} \in \mathbb{R}^{n \times n}$, $\mathbf{Q} \succ \mathbf{O}$, $\mathbf{g} \in \mathbb{R}^n$, $c \in \mathbb{R}$, $\mathbf{A} \in \mathbb{R}^{k \times n}$ with $\text{rank } \mathbf{A} = k$, and $\mathbf{b} \in \mathbb{R}^k$.

- (a). Write down the Lagrange condition for this problem.
- (b). Find a closed form solution for the optimal solution $\mathbf{x}^*$ and the corresponding Lagrange multiplier $\boldsymbol{\lambda}^*$.
Hint: Show $\mathbf{A} \mathbf{Q}^{-1} \mathbf{A}^T \succ \mathbf{O}$ and hence is invertible.
- (c). Use part (b) to find the projection $\mathcal{P}_S(\mathbf{x}_0)$ of a point $\mathbf{x}_0$ onto the affine space $S = \{\mathbf{x} : \mathbf{A} \mathbf{x} = \mathbf{b}\}$, i.e. solve

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \|\mathbf{x} - \mathbf{x}_0\|_2^2 \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \end{aligned}$$

When $\mathbf{x}_0 = \mathbf{0}$, you should recover the result on slide 11 of §9.

- (d). Consider a hyperplane $P = \{\mathbf{x} : \mathbf{w}^T \mathbf{x} = b\}$. Use the result in (c) to find the distance $\text{dist}(\mathbf{x}_0, P)$ between $\mathbf{x}_0$ and P . You should recover the result on slide 12 of §1.

Solution. (a). The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{g}^T \mathbf{x} + c + \boldsymbol{\lambda}^T (\mathbf{A} \mathbf{x} - \mathbf{b})$$

The Lagrange condition is

$$\begin{cases} \nabla_{\mathbf{x}} \mathcal{L} = \mathbf{Q} \mathbf{x} + \mathbf{g} + \mathbf{A}^T \boldsymbol{\lambda} = \mathbf{0} \\ \nabla_{\boldsymbol{\lambda}} \mathcal{L} = \mathbf{A} \mathbf{x} - \mathbf{b} = \mathbf{0} \end{cases} \quad (\text{S1})$$

- (b). From the first equation of (S1),

$$\mathbf{x} = -\mathbf{Q}^{-1}(\mathbf{g} + \mathbf{A}^T \boldsymbol{\lambda}) \quad (\text{S2})$$

Plugging into the second equation,

$$\mathbf{A} \mathbf{x} = -\mathbf{A} \mathbf{Q}^{-1}(\mathbf{g} + \mathbf{A}^T \boldsymbol{\lambda}) = \mathbf{b} \implies \boldsymbol{\lambda}^* = -[\mathbf{A} \mathbf{Q}^{-1} \mathbf{A}^T]^{-1} [\mathbf{A} \mathbf{Q}^{-1} \mathbf{g} + \mathbf{b}]$$

where we have used the fact $\mathbf{A} \mathbf{Q}^{-1} \mathbf{A}^T \succ \mathbf{O}$ and hence invertible. To see that $\mathbf{A} \mathbf{Q}^{-1} \mathbf{A}^T \succ \mathbf{O}$, first note that $\mathbf{Q}^{-1} \succ \mathbf{O}$, since $\mathbf{A}^T \mathbf{A} \succ \mathbf{O}$. Since $\mathbf{A}^T$ has full column rank, $\mathbf{y} \triangleq \mathbf{A}^T \mathbf{x} \neq \mathbf{0}$ for $\mathbf{x} \neq \mathbf{0}$. Thus for $\mathbf{x} \neq \mathbf{0}$,

$$\mathbf{x}^T \mathbf{A} \mathbf{Q}^{-1} \mathbf{A}^T \mathbf{x} = \mathbf{y}^T \mathbf{Q}^{-1} \mathbf{y} > 0$$

Plugging the expression for λ^* into (S2),

$$\mathbf{x}^* = -\mathbf{Q}^{-1}\mathbf{g} + \mathbf{Q}^{-1}\mathbf{A}^T[\mathbf{A}\mathbf{Q}^{-1}\mathbf{A}^T]^{-1}[\mathbf{A}\mathbf{Q}^{-1}\mathbf{g} + \mathbf{b}]$$

(c). Using $\mathbf{Q} = \mathbf{I}$ and $\mathbf{g} = -\mathbf{x}_0$, we obtain

$$\mathcal{P}_S(\mathbf{x}_0) = \mathbf{x}_0 - \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}(\mathbf{A}\mathbf{x}_0 - \mathbf{b})$$

(d). Use $\mathbf{A} = \mathbf{w}^T$ in (c),

$$\mathcal{P}_P(\mathbf{x}_0) = \mathbf{x}_0 - \mathbf{w}(\mathbf{w}^T\mathbf{w})^{-1}(\mathbf{w}^T\mathbf{x}_0 - b) = \mathbf{x}_0 - \frac{\mathbf{w}^T\mathbf{x}_0 - b}{\|\mathbf{w}\|^2}\mathbf{w}$$

so

$$\text{dist}(\mathbf{x}_0, P) = \|\mathbf{x}_0 - \mathcal{P}_P(\mathbf{x}_0)\| = \left\| \frac{\mathbf{w}^T\mathbf{x}_0 - b}{\|\mathbf{w}\|^2}\mathbf{w} \right\| = \frac{|\mathbf{w}^T\mathbf{x}_0 - b|}{\|\mathbf{w}\|^2}\|\mathbf{w}\| = \frac{|\mathbf{w}^T\mathbf{x}_0 - b|}{\|\mathbf{w}\|}$$

Remark. Recall on slide 12 of §1 we used the condition $\mathbf{x}_i - \mathbf{x}'_i = \gamma_i \mathbf{w}$ for some $\gamma_i \in \mathbb{R}$. Using the notation in this problem, the condition is $\mathbf{x}_0 - \mathbf{x}^* = \lambda^* \mathbf{w}$ for some $\lambda^* \in \mathbb{R}$, which is precisely the Lagrange condition for the problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2}\|\mathbf{x} - \mathbf{x}_0\|_2^2 \\ \text{s.t.} \quad & \mathbf{w}^T \mathbf{x} = b \end{aligned}$$

The Lagrange multiplier is

$$\lambda^* = \frac{\mathbf{w}^T\mathbf{x}_0 - b}{\|\mathbf{w}\|^2}$$

□

3. Solve the following problem,

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) = x_1 x_2 \\ \text{s.t.} \quad & x_1^2 + 4x_2^2 = 1 \end{aligned}$$

Proof. The global minimum exists since we are minimizing a continuous function over a nonempty compact set. Let $h(\mathbf{x}) = x_1^2 + 4x_2^2 - 1$. Then $\nabla h(\mathbf{x}) = (2x_1, 8x_2)^T \neq \mathbf{0}$ for any feasible point $\mathbf{x}$. By the Lagrange condition,

$$\begin{cases} x_2 + 2\lambda x_1 = 0 & \text{(S3a)} \\ x_1 + 8\lambda x_2 = 0 & \text{(S3b)} \\ x_1^2 + 4x_2^2 = 1 & \text{(S3c)} \end{cases}$$

By the first two equations,

$$x_2 = -2\lambda x_1 = -2\lambda(-8\lambda x_2) = 16\lambda^2 x_2$$

If $x_2 = 0$, then $x_1 = -8\lambda x_2 = 0$, violating the (S3c). Therefore, $\lambda = \pm \frac{1}{4}$. By (S3b),

$$x_1^2 = (8\lambda)^2 x_2^2 = 4x_2^2$$

Plugging into (S3c) yields $x_1 = \pm \frac{\sqrt{2}}{2}$. Thus there are four candidates.

$$\begin{cases} x_1 = \frac{\sqrt{2}}{2} \\ x_2 = -\frac{\sqrt{2}}{4} \\ \lambda = \frac{1}{4} \end{cases}, \quad \begin{cases} x_1 = -\frac{\sqrt{2}}{2} \\ x_2 = \frac{\sqrt{2}}{4} \\ \lambda = \frac{1}{4} \end{cases}, \quad \begin{cases} x_1 = \frac{\sqrt{2}}{2} \\ x_2 = \frac{\sqrt{2}}{4} \\ \lambda = -\frac{1}{4} \end{cases}, \quad \begin{cases} x_1 = -\frac{\sqrt{2}}{2} \\ x_2 = -\frac{\sqrt{2}}{4} \\ \lambda = -\frac{1}{4} \end{cases}$$

The first two solutions are the minimum, and the minimum value is $-\frac{1}{4}$. $\square$

4. In this problem, we characterize the eigenvalues of a symmetric matrix $\mathbf{A}$ as the optimal values of certain optimization problems. Let $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ be the eigenvalues of $\mathbf{A}$.

(a). Consider the following optimization problem,

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{x}^T \mathbf{A} \mathbf{x} \\ \text{s.t.} \quad & \|\mathbf{x}\|_2^2 = 1 \end{aligned} \tag{2}$$

Use the Lagrange multipliers method to show that a solution $\mathbf{x}^*$ to problem (2) is an eigenvector of $\mathbf{A}$ associated to λ_1 , and the optimal value is λ_1 .

(b). Let $\mathbf{v}_1$ be an eigenvector of $\mathbf{A}$ associated to the eigenvalue λ_1 . Consider the following optimization problem,

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{x}^T \mathbf{A} \mathbf{x} \\ \text{s.t.} \quad & \|\mathbf{x}\|_2^2 = 1 \\ & \mathbf{v}_1^T \mathbf{x} = 0 \end{aligned} \tag{3}$$

Let $\mathbf{x}^*$ be an optimal solution to (3).

i) Use the Lagrange multipliers method to show that there exist some $c_0, c_1 \in \mathbb{R}$ s.t.

$$\mathbf{A} \mathbf{x}^* = c_0 \mathbf{x}^* + c_1 \mathbf{v}_1$$

ii) Show $c_1 = 0$ in i). Hint: Consider $\mathbf{v}_1^T \mathbf{A} \mathbf{x}^*$.

iii) Conclude $\mathbf{x}^*$ is an eigenvector of $\mathbf{A}$ associated to λ_2 , and the optimal value of (3) is λ_2 . You can assume the fact that an eigenvector orthogonal to $\mathbf{v}_1$ must be associated to one of the eigenvalues $\lambda_2, \dots, \lambda_n$.

Remark. The same argument can be used to show the following generalization. Let $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$ be linearly independent eigenvectors of $\mathbf{A}$ associated to $\lambda_1, \dots, \lambda_{k-1}$, respectively. Then λ_k is the optimal value of the following problem,

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{x}^T \mathbf{A} \mathbf{x} \\ \text{s.t.} \quad & \|\mathbf{x}\|_2^2 = 1 \\ & \mathbf{v}_i^T \mathbf{x} = 0, \quad i = 1, 2, \dots, k-1 \end{aligned}$$

and the optimal solution is an eigenvector associated to λ_k .

Proof. (a). The solution exists since we are minimizing a continuous function over a nonempty compact set. Let $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ and $h(\mathbf{x}) = \mathbf{x}^T \mathbf{x} - 1$. Note that $\nabla h(\mathbf{x}) = 2\mathbf{x} \neq 0$ for any feasible $\mathbf{x}$, so all feasible $\mathbf{x}$ are regular points of h . By the Lagrange condition, there exists μ^* (we use μ^* to denote the Lagrange multiplier to avoid potential confusion with the eigenvalues λ_i) s.t.

$$\nabla f(\mathbf{x}^*) + \mu^* \nabla h(\mathbf{x}^*) = 0 \implies 2\mathbf{A} \mathbf{x}^* + \mu^* (2\mathbf{x}^*) = 0$$

So

$$\mathbf{A} \mathbf{x}^* = -\mu^* \mathbf{x}^*,$$

i.e. $\mathbf{x}^*$ is an eigenvector of $\mathbf{A}$ with associated eigenvalue $-\mu^*$. The optimal value is

$$f(\mathbf{x}^*) = (\mathbf{x}^*)^T \mathbf{A} \mathbf{x}^* = (\mathbf{x}^*)^T (-\mu^* \mathbf{x}^*) = -\mu^* \geq \lambda_1$$

The equality is achieved by an eigenvector associated to λ_1 . So $\mathbf{x}^*$ must be an eigenvector associated to λ_1 .

- (b). i) Let $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$, $h_1(\mathbf{x}) = \mathbf{x}^T \mathbf{x} - 1$, and $h_2(\mathbf{x}) = \mathbf{v}_1^T \mathbf{x}$. Note $\nabla h_1(\mathbf{x}^*) = 2\mathbf{x}^* \neq 0$ and $\nabla h_2(\mathbf{x}^*) = \mathbf{v}_1$. Since $\mathbf{v}_1^T \mathbf{x}^* = 0$, $\nabla h_1(\mathbf{x}^*)$ and $\nabla h_2(\mathbf{x}^*)$ are linearly independent, and hence $\mathbf{x}^*$ is regular. By the Lagrange condition, there exist $\mu_1^*, \mu_2^* \in \mathbb{R}$ s.t.

$$2\mathbf{A} \mathbf{x}^* + 2\mu_1^* \mathbf{x}^* + \mu_2^* \mathbf{v}_1 = 0,$$

so

$$\mathbf{A} \mathbf{x}^* = c_0 \mathbf{x}^* + c_1 \mathbf{v}_1,$$

with $c_0 = -\mu_1^*$ and $c_1 = -\frac{\mu_2^*}{2}$.

ii)

$$\mathbf{v}_1^T \mathbf{A} \mathbf{x}^* = c_0 \mathbf{v}_1^T \mathbf{x}^* + c_1 \mathbf{v}_1^T \mathbf{v}_1 = c_1 \|\mathbf{v}_1\|_2^2$$

Since $\mathbf{A} \mathbf{v}_1 = \lambda_1 \mathbf{v}_1$,

$$c_1 = \frac{\mathbf{v}_1^T \mathbf{A} \mathbf{x}^*}{\|\mathbf{v}_1\|_2^2} = \frac{\lambda_1 \mathbf{v}_1^T \mathbf{x}^*}{\|\mathbf{v}_1\|_2^2} = 0$$

- iii) $\mathbf{A} \mathbf{x}^* = c_0 \mathbf{x}^*$, so $\mathbf{x}^*$ is an eigenvector of $\mathbf{A}$ with associated eigenvalue c_0 . The optimal value is

$$f(\mathbf{x}^*) = (\mathbf{x}^*)^T \mathbf{A} \mathbf{x}^* = c_0 \|\mathbf{x}^*\|_2^2 = c_0 \geq \lambda_2$$

The equality is achieved by an eigenvector associated to λ_2 . So $\mathbf{x}^*$ must be an eigenvector associated to λ_2 . □